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Are we underestimating elasmobranch abundances on baited remote underwater video systems (BRUVS) using traditional metrics?

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ABSTRACT

Baited remote underwater video systems (BRUVS) are increasingly used to study fish communities, biomass, and animal behaviour. Due to the abundance of BRUVS data, there are many analysis methods. The most commonly used method for analysis of BRUV data is MaxN which refers to the maximum number of individuals observed of a species in a single frame of a video. Here, we present a novel method for BRUVS analysis that involves identifying and counting distinct individuals (MaxIND) to quantify the accuracy of MaxN. Individual oriental bluespotted maskray (*Neotrygon orientalis*) and the bluespotted fantail ray (*Taeniura lymma*) were identified on BRUVS by spot patterns, tail characteristics, and sex at three sites in Malaysian Borneo. We demonstrated that MaxIND gave abundances that were 2.4 and 1.1 times higher than MaxN for *N. orientalis* and *T. lymma*, respectively. These differences between methods were consistent for each species between sites regardless of the presence of marine reserves. However, differences in abundance estimates from MaxN to MaxIND were apparent between species, indicating that correction factors need to be developed on a species basis to better estimate true abundance. While identifying individuals is time consuming, it provides improved accuracy and information about populations. We therefore recommend the use of MaxIND when rare and endangered species are present, in high density populations, and for behavioural analyses.

1. Introduction

Video from Baited Remote Underwater Video Systems (BRUVS) is increasingly being used to assess fish communities and biomass (Cappo et al. 2001; Espinoza et al. 2014; Harvey et al. 2013; White et al. 2013), and animal behaviour (Hill et al. 2014; Watson et al. 2010). BRUVS collect a large amount of data and due to the abundance of data, several methods for analyses have been developed. The most commonly used technique is MaxN (Cappo 2010; Whitmarsh et al. 2016), which is a metric of species local abundance based on the maximum number of individuals observed in a single frame of video (Ebner et al. 2008; Louiseau et al. 2016). Use of MaxN as an estimate of abundance is common because it is relatively simple, fast, and easily comparable to other BRUVS analyses due to its wide use (Cappo 2010; Willis and Babcock 2000). MaxN is the most conservative estimate for total number of individuals from a species observed within a single BRUVS deployment (Whitmarsh et al. 2016) and is designed to eliminate double counting and overestimating abundance. The inability to easily distinguish between individuals of the same species on BRUVS videos means that it is not possible to assume that each appearance represents the arrival of a unique individual. As such, MaxN is likely to underestimate the true abundance of individuals in a single deployment (Kilfoil et al. 2017). Other methods of analysing BRUVS footage include mean count (MeanCount), time in – time out (TITO), and time of first arrival (T1st). MeanCount uses the number of individuals in frame at a given interval (5, 10, 30 s, etc.), to estimate abundance over time of species' presence (Cappo et al. 2011). However, MeanCount may miss individuals that pass quickly in front of the camera. TITO involves notation of the time of entry and exit of each animal included in the study, which is mostly used in behavioural analyses (Schobernd et al. 2014). Finally, T1st refers to the first entry of a species in the video, indicating the distance the animal was to the system and/or the attractiveness of the bait (Campbell et al. 2015). MaxN and MeanCount methods are used to estimate abundance and diversity of species in videos, while TITO allows for behavioural analyses like boldness using the time spent openly in view of the camera (Cappo 2010) and T1st can indicate species with better olfactory abilities based on arrival times to the BRUVS (Bassett and Montgomery 2011).

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While studies have compared BRUVS analysis methods to one another (Stobart et al. 2015), few have attempted to identify and count individuals to estimate true abundance (Harasti et al. 2016; Kilfoil et al. 2017). Previous studies have used unique markings to determine movement speed of individuals, but did not attempt to quantify the number of distinct individuals (Campbell et al. 2015; Schobernd et al. 2014). One recent study identified white shark individuals on Stereo-BRUVS, which revealed individuals were not occurring on multiple BRUVS deployed at the same time (Harasti et al. 2016). Identification of unique individuals via static photography has been used to help assess populations. Photo identification of individuals is commonly used in species with unique physical features and applied to a range of taxa including cetaceans (Evans and Hammond 2004; Thompson and Wheeler 2008), birds (Arroyo and Bretagnolle 1999; Williams and Thomson 2015) and reptiles (Bradfield 2004; Reisser et al. 2008). Photo ID has also been used extensively in elasmobranchs. For example, in white sharks (*Carcharodon carcharias*) physical features (dorsal fin markings, injuries, size, etc.) have been used to identify individuals across multiple years (Domeier and Nasby-Lucas 2007; Ryan et al. 2015). In manta rays (*Mobula* spp.) and zebra sharks (*Stegostoma fasciatum*) colour patterns and spots unique to each individual have been used for identification of individuals (Dudgeon et al. 2008; Germanov and Marshall 2014). In BRUVS footage, multiple angles of an individual are often seen enabling detection of distinguishing features making identification of individuals possible in some species.

Surveying batoid (rays, skates, and guitarfish) populations is challenging due to their often cryptic nature and caution around larger animals, including humans (Cappo et al. 2001; Espinoza et al. 2014; Harvey et al. 2013; White et al. 2013). Currently, fishing surveys are the most commonly used method to estimate batoid population abundances and how they change through time. Fisheries sampling bias through targeting desirable species, preferentially fishing in certain areas, gear selectivity, and data limitations mean that catch data may not adequately represent batoid diversity (Graham et al. 2001; Walker and Hislop 1998). However, emerging methodologies such as fishery-independent BRUVS can overcome some of these issues, and are a low impact means of sampling batoid populations (White et al. 2013). As many batoid populations are currently decreasing at a rapid rate globally and are also poorly studied (Dulvy et al. 2014), there is a need for accurate abundance estimates to help inform management and conservation efforts. The aims of this study were to: a) determine if individual batoids could be distinguished in BRUVS footage, and b) examine differences in MaxN compared to results from counts of identified individuals. It was expected that species with unique markings would be able to be distinguished using BRUVS footage and MaxN would significantly underestimate the true abundance.

2. Methods

2.1. Study site

Three sites in Malaysian Borneo (Tunku Abdul Rahman Park (TARP), Tun Sakaran Marine Park (TSMP) and the islands of Mabul and Kapalai (MK)) were sampled with BRUVS. All sites consisted of patchy coral reefs with varying degrees of reef degradation within each site. The TARP (5°59′22.06″N, 116°1′25.28″E), established in 1974, is located 3 km off the coast of Kota Kinabalu and consists of five islands over an area of 49 km^2: Gaya, *Sapi*, Mamutik, Manukan, and Sulug. The TARP is closed to fishing, however many recreational water activities occur in the park such as SCUBA diving, snorkelling, and parasailing. The TSMP (4°38′21.52″N, 118°44′0.13″E) is located 18 km northeast of Semporna and was established in 2004. The TSMP has an area of 101 km^2, and consists of seven islands and one patch reef: Boheydulong, Bodgaya, Sabangkat, Salakan, Maiga, Sibuan, Mantabuan, and Church Reef. The TSMP is restricted to subsistence fishing, however, the enforcement level is low (Sherman pers. obs.). The main activities in the TSMP include scuba diving and snorkelling. Mabul and Kapalai (4°13′49.12″N, 118°39′19.55″E) are located 25 km south of Semporna and consist of an area ~20 km^2. Both islands are open to fishing, however, they are mainly used for SCUBA diving with > 25 operators in the area. Subsistence fishing occurs daily with occasional trawlers operating within 1 km of the islands.

2.2. Sampling

Baited remote underwater video systems (BRUVS) were deployed during daylight hours in a variety of habitats including fore reef, reef crest, reef flat, and lagoon at depths from 1.5 m to 40 m. BRUVS used in this study consisted of aluminium frames that housed a GoPro Hero 4 Silver camera with wide angle view (approx. 170° in air), (1920 × 1080 video format, 30 frames/s) housed in NiMAR housings, and a bait arm that extended 1 m from the camera. The bait arm held a mesh bag containing approximately 1 kg of crushed pilchards (*Sardinella* spp.) or slimy mackerel (*Scomber australiasicus*). BRUVS were manually lowered to the seafloor and recovered using floating rope attached to a surface buoy marking the location. Six BRUVS were deployed at one time with each BRUV left to record video footage throughout a minimum 60 min deployment period. BRUVS were deployed with a minimum of 500 m between each BRUVS, a distance at which it was assumed that batoids would not swim between adjacent cameras within the deployment period. Up to 24 BRUVS were set in a single day through multiple tidal states with fresh bait used for each deployment. During deployments the boat maintained a distance of at least 200 m to reduce any effects of boat noise on animal behaviour.

2.3. Video annotation

All BRUVS footage was watched by two independent, trained annotators using Event Measure software (www.seagis.com v.4.43). Annotators marked the arrival time of every batoid that entered the screen throughout the video. A senior reviewer validated species identification and compared the two reads of each video. If the two reads differed, a third independent annotator was used to determine which of the first two reads was correct. Of 286 videos, 11 required a third reader. In all 11 cases, the third reader's results matched one of the first two annotations, therefore this was deemed the final annotation. This indicated consistent and reliable results from the trained annotators.

2.4. Species

Two batoid species were examined for this study; the oriental bluespotted maskray (*Neotrygon orientalis* Last, White & Séret 2016), and the bluespotted fantail ray (*Taeniura lymma*Forsskål, 1775). These species were selected because they were the two most frequently observed batoids on Malaysian BRUVS.

2.5. Individual identification

To investigate how many different individuals were present, all videos with *N. orientalis* and *T. lymma* were reanalysed. Each time a batoid was within the field of view, the best possible frames were extracted from the video to illustrate key identifying features (Fig. 1). Frames were then compared to differentiate between individuals within each deployment. When individuals were not identifiable (too distant or moving too quickly) they were labelled as “unknown.” No studies have been performed to determine the longevity and reliability of re-identification using markings of either species of ray in this study. However, the maximum time in which the rays could be re-identified was 90 min, therefore it was concluded that these features would not change in this time period. No attempt was made to identify individuals across deployments.

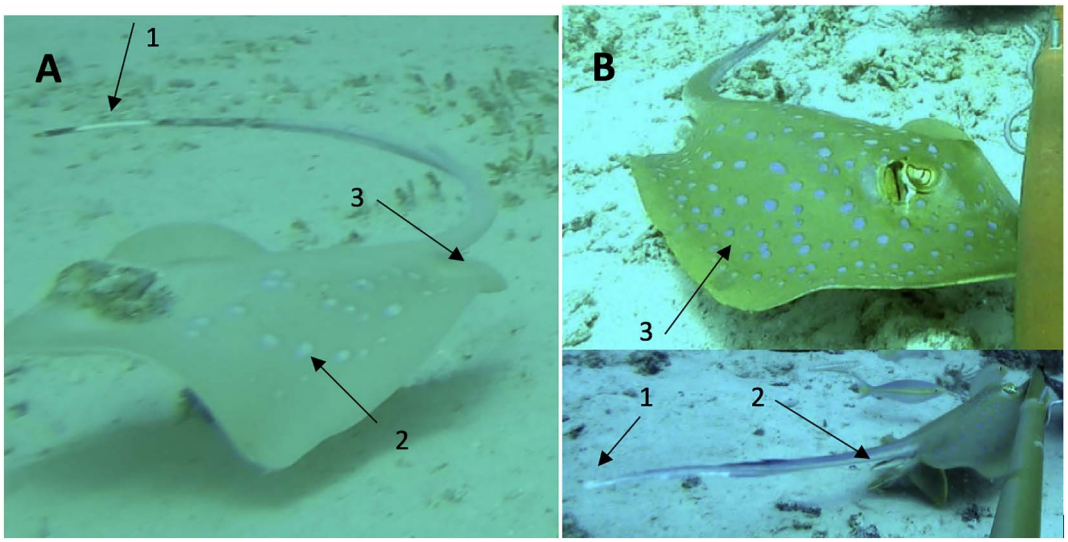

Fig. 1. Features used to distinguish individuals of the (A) oriental bluespotted maskray: (1) "Barcode" – number of iterations of black and white, size of segments, presence/absence, (2) Spot patterns – clusters of spots, left and/or right pectoral fins, (3) Sex – male/female and (B) bluespotted fantail ray (1) Tail scratches/ bites – small imperfections in tail flap and white tip of tail, (2) Sex – male/female, (3) Spot patterns – clusters of spots, left and/or right pectoral fins, rostrum. Where possible, multiple features were used in conjunction for distinguishing individuals.

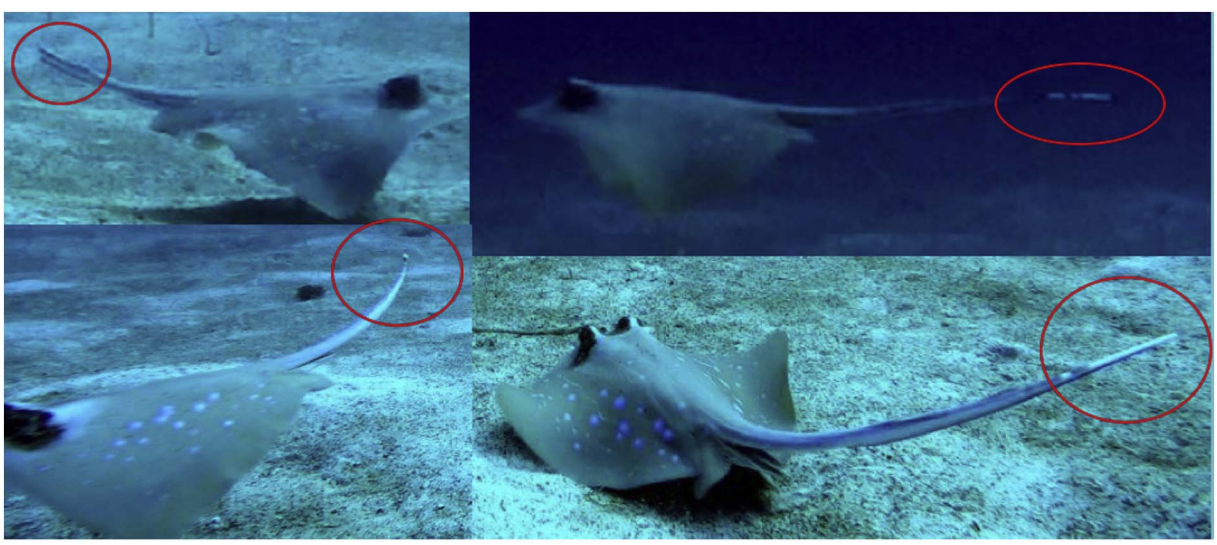

Fig. 2. Four oriental bluespotted maskray individuals from the same video. This video had a MaxN of one, 11 rays were observed in front of the camera and four individuals were distinguished through differences in their tails.

Individual *N. orientalis* were most readily identified by their unique "barcode" markings on their tails and secondarily by spot patterns on the pectoral fins (Fig. 2). In *T. lymma* tail imperfections and sex were most frequently used to distinguish individuals. Spot patterns were rarely used for *T. lymma* as the blue spots were visible only when an individual was slowly moving close to the camera which did not occur often. *Neotrygon orientalis* individuals were more easily distinguished than *T. lymma* because they had more easily distinguishable tails.

2.6. Analyses

Paired *t*-tests (R v0.99.491) were used to determine differences in number of times a ray was in frame during a deployment and the number of occurrences of distinguishable individuals for both species. Additionally, paired t-tests were used to determine if the MaxN from each video was significantly different than the number of individuals identified (MaxIND) using morphological traits for both species. Two-way ANOVA was used to determine if there was a site or species effect on the proportion of sightings of distinguishable individuals.

3. Results

A total of 312 BRUVS were deployed throughout this study. Due to low visibility (< 1 m), 26 videos were removed from analysis leaving 97 from TARP, 96 from TSMP, and 67 from MK for a total of 286 successful BRUVS. From these 286 deployments, 372 occurrences of *Neotrygon orientalis* and *Taeniura lymma* were recorded. Overall, *T. lymma* were present in more videos (22.6%) than *N. orientalis* (12.3%).

From 372 occurrences of these two ray species, 282 could be identified to the individual. In *N. orientalis* 64 occurrences were made by individuals only appearing once, 22 individuals passed through the frame twice in a video, and 17 individuals occurred 3+ times in a single video, for a total of 103 unique individuals identified. In 55 cases *T. lymma* individuals occurred once in a video, 7 individuals occurred twice, and 5 individuals occurred 3+ times in a single video for a total of 67 unique individuals identified. A significant number of rays

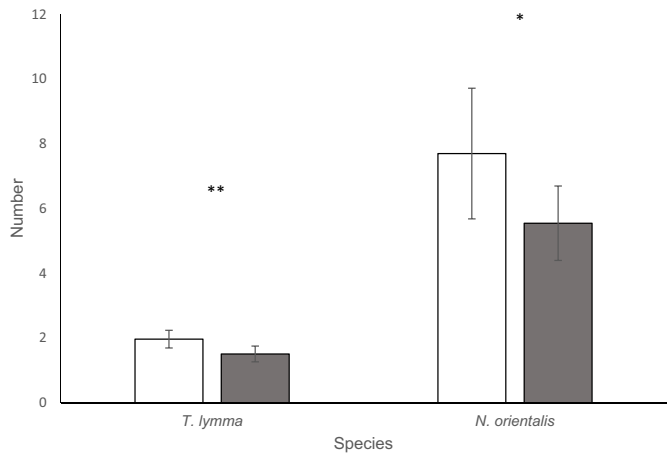

Fig. 3. Mean numbers of identified passes by an individual in front of the camera was significantly lower than the overall number of rays that passed through the field of view in each video for both species (* $p < 0.05$; ** $p < 0.0001$). Error bars represent one standard deviation, shaded bars represent number of identified passes and white bars represent total number of total times an individual of the species was in the field of view.

moving through the frame were not able to be distinguished from other individuals for both species (*N. orientalis*: $t = 2.24$, df = 32, $p = 0.0323$; *T. lymma*: $t = 4.69$, df = 58, $p < 0.0001$) (Fig. 3).

A MaxN of two and three occurred 12.1% and 9.1% of the time, respectively. Throughout the three sites, *T. lymma* occurred on more deployments, but in lower numbers than *N. orientalis*. In 96.6% of videos the MaxN of *T. lymma* was one (a single individual present in the frame). Two videos had a MaxN of two. *Neotrygon orientalis* had a MaxN of one 78.8% of the time.

Use of MaxIND resulted in significantly greater numbers of both species being recorded than using MaxN (*N. orientalis:* $t = 3.59$, df = 32, $p = 0.0010$; *T. lymma:* $t = 2.19$, df = 58, $p = 0.0327$) (Fig. 4). Overall, MaxIND estimates were 2.4 X greater than MaxN from all videos for *N. orientalis* (Fig. 5) and 1.1× greater for *T. lymma.* Time between re-sighting of the same individual ranged from 1 s to 22.1 min in *N. orientalis* and 1 s to 49.7 min for *T. lymma.* There was also no significant difference in the proportion of identified individuals of the two species or among the three sites (*N. orientalis:* $F_{2,85} = 0.3730$, $p = 0.5430$; *T. lymma:* $F_{2,85} = 0.0293$, $p = 0.9711$) (Table 1). The largest difference in a single deployment occurred for *N. orientalis*, which had a MaxN of 3 and a MaxIND of 17.

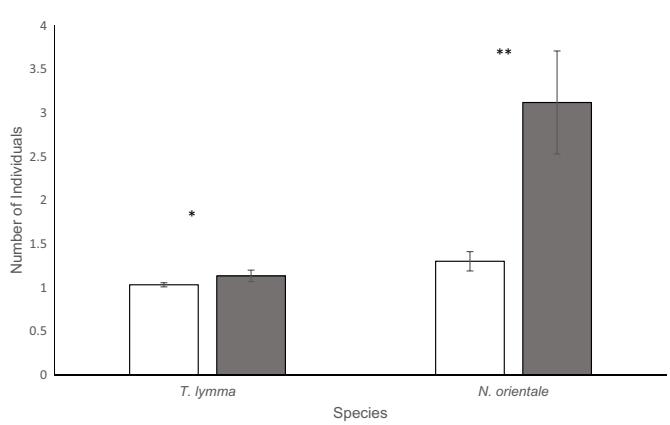

Fig. 4. Mean numbers of identified *T. lymma* and *N. orientalis* were significantly lower than the mean number of passes made by the species (* $p < 0.05$; ** $p < 0.005$). Error bars represent one standard deviation, shaded bars represent number of identified individuals (MaxIND) and white bars represent MaxN.

4. Discussion

The results of this study quantified the potential for abundance underestimation in the most widely used metric of relative abundance in analysis of BRUVS data, indicating that MaxN can underestimate the true values of abundance by 2.4× for some species. While MaxN is commonly used for abundance estimates as it is simple and fast (Cappo 2010), it includes uncertainty when individuals swim in and out of frame. In these cases, it may be a single individual circling or multiple individuals swimming past. In contrast, counting each individual that enters as a new individual could substantially overestimate true abundance. Using MaxN removes the risk of overestimation as only the maximum number of individuals in a single frame are counted. Therefore trade-offs in analyses and outcomes need to be considered when selecting analysis methods.

BRUVS studies can be used to help detect marine reserve effects (Espinoza et al. 2014; Goetze and Fullwood 2012; Hill et al. 2014; Schobernd et al. 2014; Whitmarsh et al. 2016). We have shown that differences between MaxN and MaxIND were consistent for both species in this study between three different sites, including two within marine reserves. Therefore, conclusions drawn about marine reserve effectiveness in previous studies that have used MaxN remain valid even though abundance estimates have likely been underestimated.

There was no significant difference in abundance underestimation between sites, but differences between species were apparent. Abundance estimates from MaxIND were higher than MaxN by 1.1× for *Taeniura lymma* and 2.4× for *Neotrygon orientalis* revealing stark differences in outputs and conclusions about population size. However, MaxIND requires the capacity to identify unique individuals via some physical feature to produce species-specific estimates. This means that the MaxIND method is not suitable for all species, and that MaxN is a more appropriate method for analysis when individuals cannot be identified. The MaxIND method would best be used as a supplement to MaxN for species where individual identification is possible.

A further consideration is that identification of individuals requires more analysis time than MaxN alone. However, new technology such as pattern recognition and identification software may be able to reduce this time discrepancy (Siddiqui et al. 2017), and indeed, is being used for a range of sharks and rays (Marshall and Pierce 2012). Increasing camera field of view angles may also improve estimates of abundance and recognition of individuals by reducing the time 'out of frame'. For example, in some cases *T. lymma* individuals would appear early in the video and return up to 50 min later. Since the GoPro™ cameras were set with only a 170° field of view in a single direction there is no way to know how far individuals travelled when out of view before returning to the BRUVS and re-entering the camera's field of view. Use of 360° cameras may increase accuracy of abundance estimates by monitoring a greater area around the BRUV and reduce loss of individuals moving in and out of frame. 360° cameras are beginning to be used for underwater surveys to examine habitat or fish assemblages (Sanguinetti 2013; Taylor et al. 2013). They have already confirmed MaxN can significantly underestimate abundance, particularly in high density populations (Kilfoil et al. 2017). However, use of 360° cameras also increases costs and analysis times.

As no studies have been performed to determine the longevity and reliability of unique markings used to re-identify the species of ray in this study, distinguishing individuals beyond single deployments was not possible. In other batoid species, like the spotted eagle ray (*Aetobatus narinari*) and manta rays (*Mobula alfredi* or *birostris*) spot patterns have been shown to be consistent over time (González-Ramos et al. 2016; Marshall et al. 2011). A mark-recapture study or aquaria study would need to be performed to determine if the "barcode" on *N. orientalis* is stable throughout their life. Using tail imperfections of *T. lymma* would likely not be as feasible as differences were subtle and will change throughout their lives. Stingrays have a well-documented ability to shed and regrow their spines (Johansson et al. 2004; Lowe

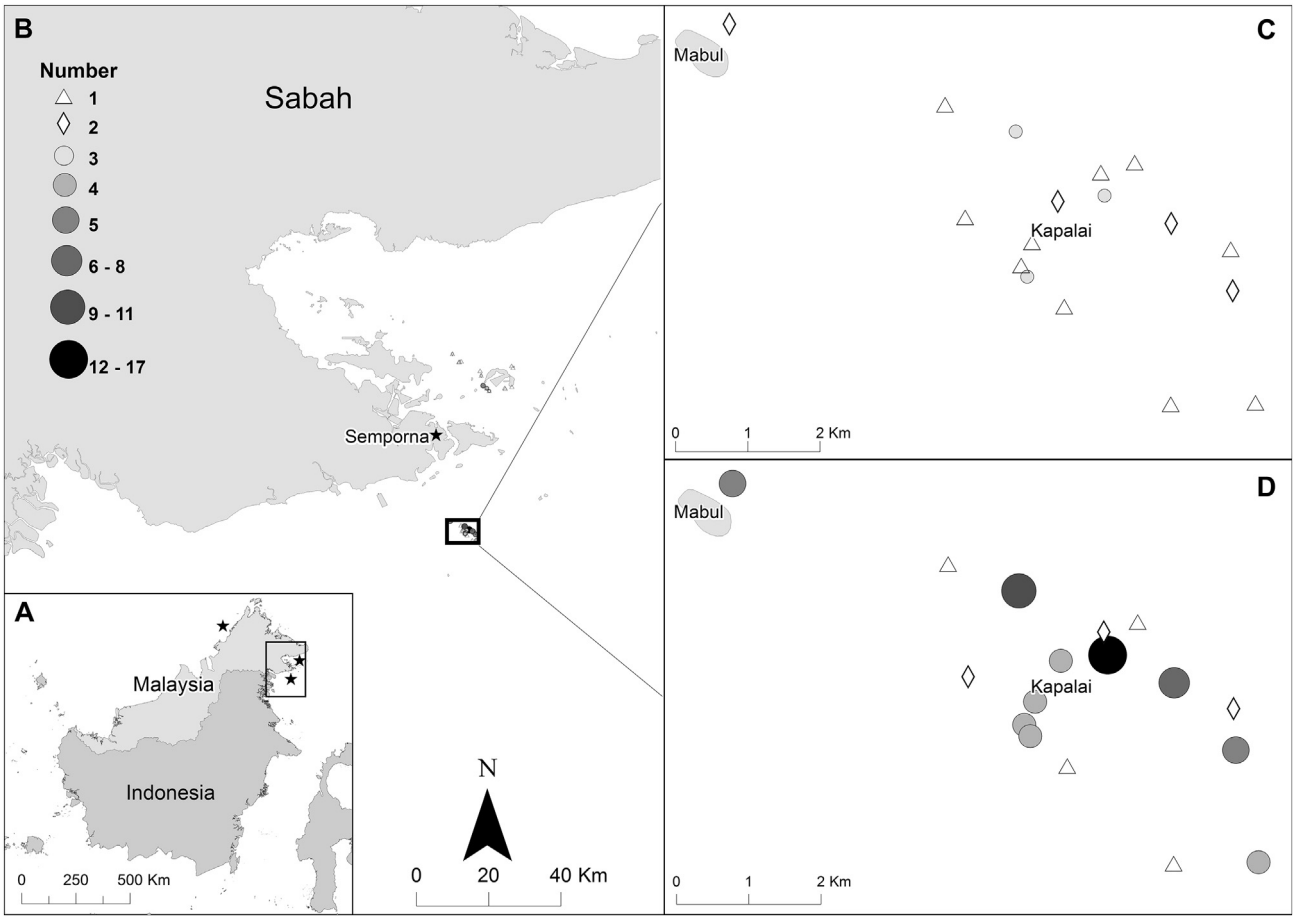

Fig. 5. A) Borneo. Stars indicate three sampling sites in Malaysian Borneo. B) BRUVS drops off Semporna. C) MaxN of drops with oriental bluespotted maskrays present. Only videos with at least one maskray present were included. D) Number of identified individuals on BRUVS with oriental bluespotted maskrays present.

Table 1
Neotrygon orientalis and *Taeniura lymma* observations and identification at each of the three sites in this study.

Species	*Neotrygon orientalis*				*Taeniura lymma*			
Site	TSMP	TARP	MK	Total	TSMP	TARP	MK	Total
Videos present	13	2	17	32	32	19	8	59
Videos with a single pass	2	2	4	8	17	14	7	38
Total observations	58	2	193	253	76	30	10	116
Identifiable individuals	51	2	129	182	59	22	8	89

et al. 2007), however, the healing and growth of the rest of their tail is currently unknown. If persistent distinguishing features could be identified, individuals could be counted beyond single deployments and would enable additional questions to be investigated including determining whether individuals move between BRUVS deployments and are counted multiple times, or longer term and larger scale movement patterns. Another application would be in behavioural studies, as repeated entries by the same individual can be analysed. In most current methodologies BRUVS are assumed to be set sufficiently far apart for individuals not to be recounted. Identification of individuals would facilitate analyses of foraging distances and swimming speed. When deploying BRUVS over multiple days, re-identification of individuals would also facilitate population estimates similar to mark recapture studies (Dempster et al. 2010; Koenig and Stallings 2015; Lee et al. 2015; Pelletier et al. 2012).

Although MaxIND requires more analysis time than MaxN, it's improved accuracy may be advantageous in some studies such as those focusing on rare and/or endangered species, surveying high density populations, and exploring individual behaviours. Populations of rare and/or endangered species are difficult to quantify due to zero-inflated data sets (Cunningham and Lindenmayer 2005). Using MaxIND to assess and monitor these species can provide more accurate abundance estimates for populations where small differences may have great significance in management and conservation (McConville et al. 2009). Finally, MaxIND, enables observation of behaviours for individuals that repeatedly enter and exit the frame. This allows for longer observation time of each individual and may even show dominance hierarchies when multiple individuals are present simultaneously (Nakano 1995). Use of 360° cameras has also shown that species occurring in high densities are often significantly undercounted using single camera BRUVS (Kilfoil et al. 2017). Using MaxIND reduces the risk of undercounting in high density areas, where a MaxN threshold may occur. Additionally, repeated sampling using MaxIND would better show population changes over time as the abundance estimates would be more accurate.

5. Conclusions

The identification of unique individuals on BRUVS deployments demonstrated that conventional BRUVS abundance metrics may underestimate the true abundance of individuals in a single deployment. By identifying individuals through unique spot patterns, tail markings, and sex, we determined that abundance of *Neotrygon orientalis* and *Taeniura lymma* were underestimated by MaxN. Although identifying individuals achieved an estimate closer to true abundance, the time for

analysis was much greater. Therefore the benefits of identifying individuals and counting MaxIND must be considered relative to the amount of time and effort required to collect the information and the specific research questions being addressed. Identifying individuals, although time consuming, is a valuable method and provides the opportunity to address a larger suite of questions than when applying MaxN alone. MaxIND is a useful tool when dealing with rare and/or endangered species as it will provide more accurate abundances of these species. MaxIND will also be useful for species that can be found in groups as the entire group may not be observed in the frame at a given time but counting distinct individuals will accurately reflect the abundance.

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